

EXP NO : 05

DATE :

VERSION CONTROL SYSTEM SETUP AND USAGE USING GIT

Aim:

Version Control System setup and usage using GIT.

- Creating a repository
- Checking out a repository
- Adding content to the repository
- Committing the data to a repository
- Updating the local copy
- Comparing different revisions
- Revert
- Conflicts and solving a conflict

Program:

```
git init
git status

echo "Hello Git" > 1.txt
echo "This is version control system setup lab." > 2.txt

git status
git add 1.txt 2.txt
git commit -m "Initial commit with 1.txt and 2.txt"

echo "New line added to 1.txt" >> 1.txt
git diff
git add 1.txt
git commit -m "Updated 1.txt with new line"

git log --oneline
git diff HEAD~1 HEAD
git revert --no-edit HEAD

git branch test-branch
git checkout test-branch
echo "Edit in test-branch" >> 1.txt
git commit -am "Change in test-branch"

git checkout main
echo "Edit in main branch" >> 1.txt
git commit -am "Change in main branch"

git merge test-branch || echo ">>> Merge conflict created. Resolve 1.txt manually."

echo ">>> If conflict occurs, open 1.txt, fix it, then run:"
echo "  git add 1.txt && git commit -m 'Resolved merge conflict in 1.txt'"
```

Output:

```
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen$ cd P122
-bash: cd: P122: No such file or directory
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen$ mkdir P122
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen$ cd P122
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen/P122$ git init
git status
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:     git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:     git branch -m <name>
Initialized empty Git repository in /mnt/c/Users/praveen/P122/.git/
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen/P122$ echo "Hello Git" > 1.txt
echo "This is version control system setup lab." > 2.txt
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen/P122$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    1.txt
    2.txt

nothing to commit but untracked files present (use "git add" to track)
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen/P122$ |
```

```
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen/P122$ git log --oneline
git diff HEAD~1 HEAD
git revert --no-edit HEAD
5f5e6e7 (HEAD -> master) Updated 1.txt with new line
8e85f86 Initial commit with 1.txt and 2.txt
diff --git a/1.txt b/1.txt
index 9f4d96d..c59b26c 100644
--- a/1.txt
+++ b/1.txt
@@ -1,2 @@
 Hello Git
+New line added to 1.txt
[master: 8db9029] Revert "Updated 1.txt with new line"
Date: Mon Aug 25 02:54:26 2025 +0000
1 file changed, 1 deletion(-)
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen/P122$
git branch test-branch
git checkout test-branch
echo "Edit in test-branch" >> 1.txt
git commit -am "Change in test-branch"
Switched to branch 'test-branch'
[test-branch: 0b2572d] Change in test-branch
1 file changed, 1 insertion(+)
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen/P122$ git checkout main
error: pathspec 'main' did not match any file(s) known to git
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen/P122$ echo "Edit in main branch" >> 1.txt
git commit -am "Change in main branch"
[test-branch: 09b0e08] Change in main branch
1 file changed, 1 insertion(+)
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen/P122$
```

```
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen/P122$ git add 1.txt 2.txt
git commit -m "Initial commit with 1.txt and 2.txt"
[master (root-commit) 8e85f86] Initial commit with 1.txt and 2.txt
2 files changed, 2 insertions(+)
create mode 100644 1.txt
create mode 100644 2.txt
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen/P122$ echo "New line added to 1.txt" >> 1.txt
git diff
git add 1.txt
git commit -m "Updated 1.txt with new line"
diff --git a/1.txt b/1.txt
index 9f4d96d..c59b26c 100644
--- a/1.txt
+++ b/1.txt
@@ -1,2 @@
 Hello Git
+New line added to 1.txt
[master 5f5e6e7] Updated 1.txt with new line
1 file changed, 1 insertion(+)
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen/P122$ git log --oneline
git diff HEAD~1 HEAD
git revert --no-edit HEAD
5f5e6e7 (HEAD -> master) Updated 1.txt with new line
8e85f86 Initial commit with 1.txt and 2.txt
diff --git a/1.txt b/1.txt
index 9f4d96d..c59b26c 100644
--- a/1.txt
+++ b/1.txt
@@ -1,2 @@
 Hello Git
+New line added to 1.txt
[master 8db9029] Revert "Updated 1.txt with new line"
Date: Mon Aug 25 02:54:26 2025 +0000
1 file changed, 1 deletion(-)
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen/P122$ |
```

```
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen/P122$ git checkout main
error: pathspec 'main' did not match any file(s) known to git
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen/P122$ echo "Edit in main branch" >> 1.txt
git commit -am "Change in main branch"
[test-branch: 09b0e08] Change in main branch
1 file changed, 1 insertion(+)
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen/P122$ git merge test-branch || echo ">>> Merge conflict created. Resolve
echo ">>> If conflict occurs, open 1.txt, fix it, then run:"
echo "    git add 1.txt && git commit -m 'Resolved merge conflict in 1.txt'"
Already up to date.
>>> If conflict occurs, open 1.txt, fix it, then run:
    git add 1.txt && git commit -m 'Resolved merge conflict in 1.txt'
praveen@LAPTOP-MDD7N44U:/mnt/c/Users/praveen/P122$ |
```

Result:

Thus, the Version Control System was set up and used successfully using Git. The commands were executed in a bash file, and the output was verified.